

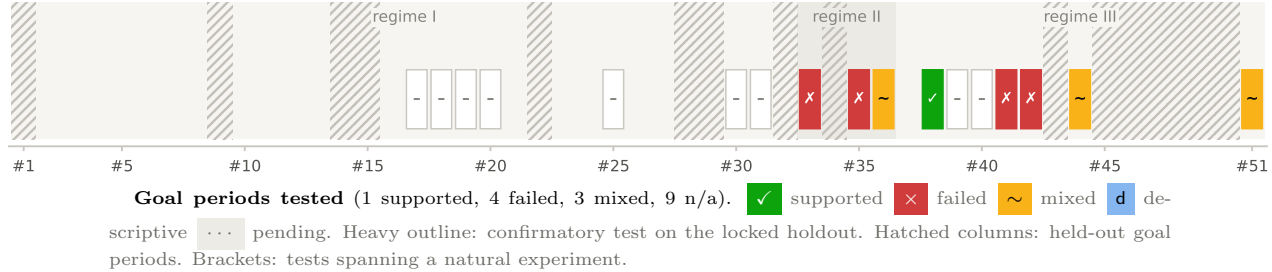
H33: The diversity–productivity curve is an inverted U: the swarm’s operating point

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 0.5/5 Usefulness 1.5/5

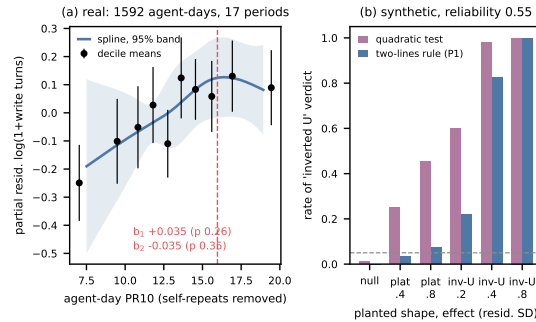
Bottom line. Agent-day content diversity has no output optimum in this swarm, measured in write turns or in commits that landed: the curve is flat, curvature hurts out of sample, and the only right-side drop is a #51 property that one embedding model does not see.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Does an agent’s daily content diversity have an output optimum? For each agent-day: the participation ratio PR10 of its chat (self near-copies removed, rarefied to 10 statements, whitened bge) against $\log(1 + \text{write turns})$, within agent and period, with day effects and activity controls. 17 non-holdout periods with logged write verbs, 1592 agent-days. The primary test is Simonsohn’s two-lines test, validated first on the real design: 0–7% false positives, 82% power for a 0.4-SD inverted U. **No inverted U:** the slope is $+0.035$ ($p = 0.26$) below the breakpoint and -0.035 ($p = 0.35$) above it, and the curve is flat from PR10 ≈ 14 up. Per period: 0 supported, 3 mixed, 11 failed, and the meta-analysed right slope has the wrong sign. PR10 improves day-blocked CV by 0.2%; adding curvature makes it worse. Self-repetition does not predict same-day output either ($p = 0.84$). **Round 1b** (work commits, #30+, both embedding models): the null stands. Holdout script frozen, not run.



Observables. (a) Partial residuals of $\log(1 + \text{write turns})$ by PR10 decile (within agent×period, day effects and activity controls removed) with the 4-df spline: a gentle, non-significant rise that flattens above PR10 ≈ 14 . Dashed line: two-lines breakpoint. (b) Synthetic replay on the real design at PR10’s measured reliability (0.55). The two-lines rule rarely calls a plateau an inverted U; the quadratic test does so 25–46% of the time. The rule detects a 0.4-SD inverted U 82% of the time.

PHYSICAL MODEL

Model 11 (vector spins). Each chat statement is a unit vector in the regime’s whitened 32-d bge space. An agent-day’s covariance Σ has participation ratio PR: 1 is a frozen spin repeating itself, 32 is isotropic (“paramagnetic”) content. H33 claims output peaks at an intermediate PR. Output: write turns (git push or commit, deploy, PR/MR create or merge) per agent-day, $y_{id} = \log(1 + \text{writes})$; α_{iu} , δ_d : agent×period and day effects; z : activity controls; x_c : Robin Hood breakpoint; $\nu = n - 1$, $n = 10$.

$$\text{PR} = (\text{tr } \Sigma)^2 / \text{tr } \Sigma^2 \quad (\hat{a}, \hat{b} \text{ unbiased from } W = \nu S)$$

$$y_{id} = \alpha_{iu} + \delta_d + f(\text{PR}_{id}) + \gamma^\top z_{id} + \varepsilon_{id}$$

$$f = b_1(x - x_c)[x < x_c] + b_2(x - x_c)[x \geq x_c] + c[x \geq x_c]$$

inverted U: $b_1 > 0$, $b_2 < 0$ ($p < .05$), $\arg \max f$ interior

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For operators: there is no content-diversity operating point to steer toward. Pushing agents to talk about more things, or fewer, should not be expected to change how much they ship. Write turns at the best PR10 and at the 90th percentile differ by at most about 9%. The only hint is on the loop side (the bottom decile may write about 25% less, not significantly), which H12’s self-repetition rate catches more directly. Caveats: daily chat PR is half noise, and write turns count activity, not value. A clean null that retires a plausible dashboard metric.

RESULTS VS PREDICTIONS

Prediction	Round 1 (write turns) → 1b Verdict (work commits)
P1 inverted U	$b_2 -0.035$ ($p .35$) → -0.040 ($p .31$); $\times$ b_1 n.s.
P2 per period	meta $b_2 +0.028$ → $+0.017$ (wrong $\times$ sign); 1/8 supported
P3 curvature in CV	spline worse than linear (5/5 seeds) $\times$ both rounds
P6 loops cost out-put	$+0.17$ → $+0.97$ (n.s.); $-$ in #31, $\times$ #35, #42
P7 swarm-day	$b_1 -0.22$ → -0.25 $\times$
gte / dedups (new)	$b_2 < 0$ in all 8 variants; sig. in 3 bge #51 specs, none gte; 0 without #51 tail
G51 rival pairs (native)	same role, same day: quad $p = 0.58$ null tail
G51 role classes (native)	support class (4 agents) concave, $p < .001$ lead
G39 own worlds (native)	$\rho(\text{PR10, commits}) +0.10$ null
G42 videos (native)	$\rho(\text{PR10, videos}) +0.10$ null

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

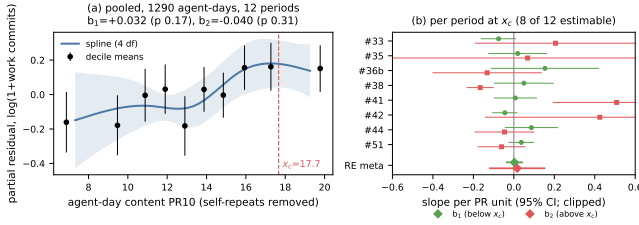
CAVEATS

The work ledger is dense only from #30, so 5 round-1 periods drop out and #51 is 58% of the pool. Agent-day PR10 is moderately model-dependent (bge vs gte $\rho = 0.77$) and half noise (reliability 0.55). The right-side drop appears in some bge specifications only and vanishes without #51. The support-class inverted U rests on 4 agents. Commits count landed work, not value.

ROUND 2

Leave as a closed null unless the #51 tail lead (C5) is confirmed on the holdout; if anything, test the support-role curve on the #51 tail. Hour-scale designs and Jev progress scores are the only untried angles.

No round-2 redirects yet.



Second figure. Round 1b: (a) partial residuals of $\log(1 + \text{work commits})$ by PR10 decile, 12 eligible periods from #30 on, 1,290 agent-days, with the 4-df spline and the two-lines breakpoint; (b) per-period slopes below and above the breakpoint with the random-effects meta (the right slope's meta has the wrong sign).

GOAL PERIODS IN DETAIL

- G33 $\times$ 1b: failed (work commits: $b_1 -0.076$, $b_2 +0.206$ ($p 0.3...$))
- G35 $\times$ 1b: failed (work commits: $b_1 +0.019$, $b_2 +0.068$ ($p 0.9...$))
- G36 $\sim$ 1b: mixed (work commits: $b_1 +0.154$, $b_2 -0.132$ ($p 0.36...$))
- G38 $\checkmark$ 1b: supported (work commits: $b_1 +0.050$, $b_2 -0.166$ ($p ...$))
- G41 $\times$ 1b: failed (work commits: $b_1 +0.007$, $b_2 +0.508$ ($p 0.0...$))
- G42 $\times$ 1b: failed (work commits: $b_1 -0.044$, $b_2 +0.425$ ($p 0.1...$))
- G44 $\sim$ 1b: mixed (work commits: $b_1 +0.087$, $b_2 -0.046$ ($p 0.54...$))
- G51 $\sim$ 1b: mixed (work commits: $b_1 +0.037$, $b_2 -0.061$ ($p 0.32...$))
- G17 $-$ 1b: n/a (pre-#30: work-ledger zeros ambiguous; 2026-1...
- G18 $-$ 1b: n/a (pre-#30: work-ledger zeros ambiguous; 2026-1...

+7 more periods: see the card.